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KAIST

Photo-doping of Graphene Enhanced by Stable Perovskite and Hole Transport Layer

Sang Hoon Kim[1], Sergey Menabde[1], and Min Seok Jang[1*]

Speaker:
Sang Hoon Kim

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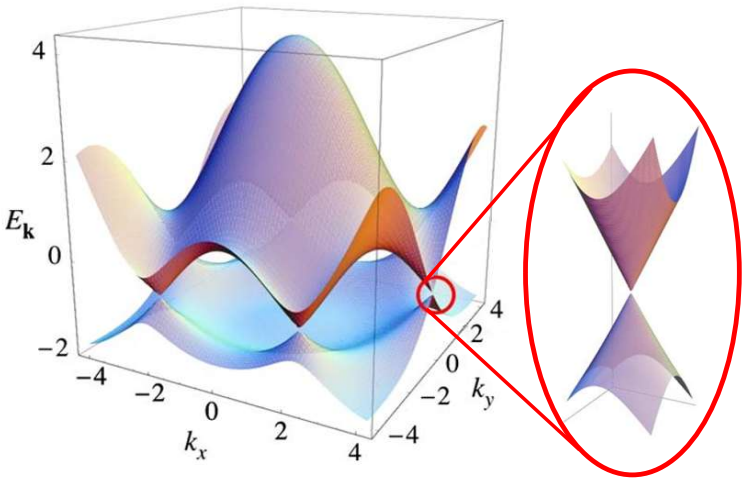
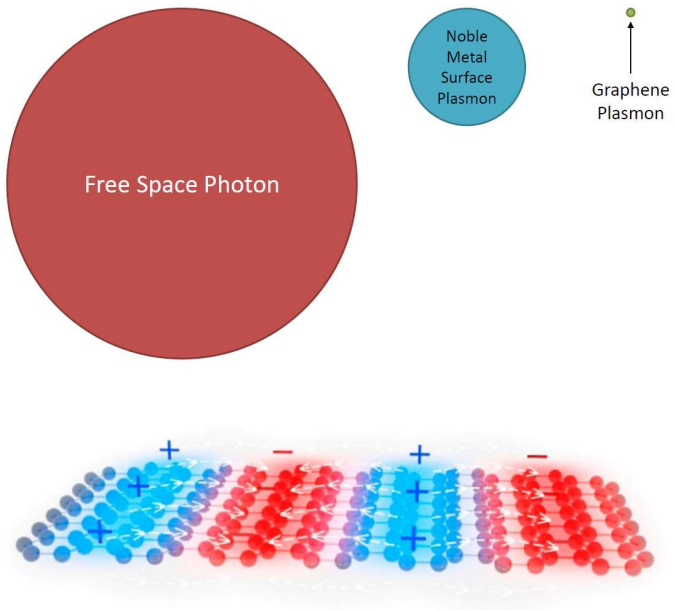
Introduction: Graphene Optoelectronics

Zero-bandgap, Linear DOS
Ultra high mobility



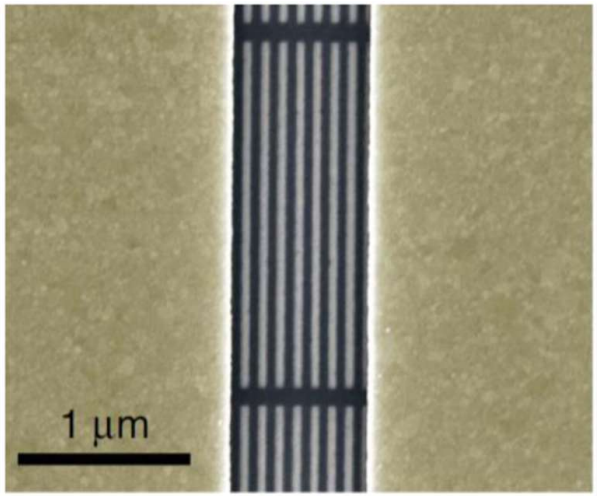
- $\rho(E) \approx \frac{2}{\pi} \cdot \frac{|E|}{(\hbar v_F)^2}$
- $E_F \approx \hbar v_f \sqrt{\pi p}$

Highly confined graphene plasmon



- $\lambda_p \sim \frac{\lambda_0}{100}$

Easily controlled plasmon frequency



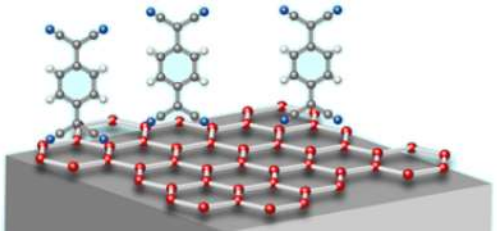
- $\hbar\omega_p = 4e\sqrt{\frac{1}{\pi}\frac{2}{\varepsilon_1+\varepsilon_2}}\sqrt{\frac{E_F}{W}}$

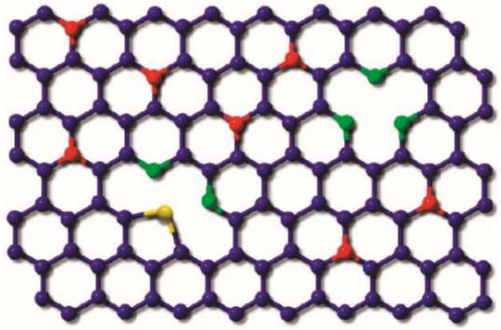
AHC Neto et al., *Rev. Mod. Phys.*, 2009, 81, 109.
FJ Garcia de Abajo, *ACS Photonics*, 2014, 1, 135-152.
Q. Guo et al., *ACS Photonics,* 2017, 4, 2989-2999.
S. Kim et al., *Nature communications,* 2016, 7, 12323.

Graphene Doping

Chemical doping

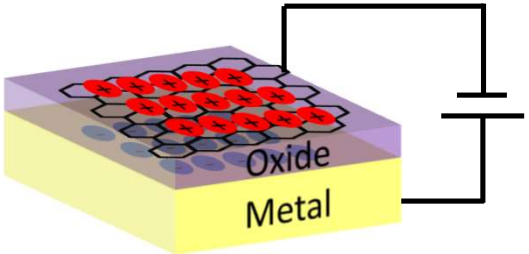
Pros: Permanent doping
Cons: Passive control





Electrical doping

Pros: Dynamic control
Cons: Fixed gating pattern



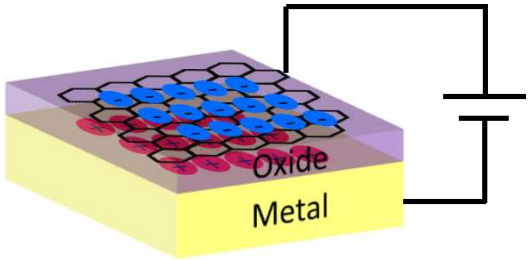
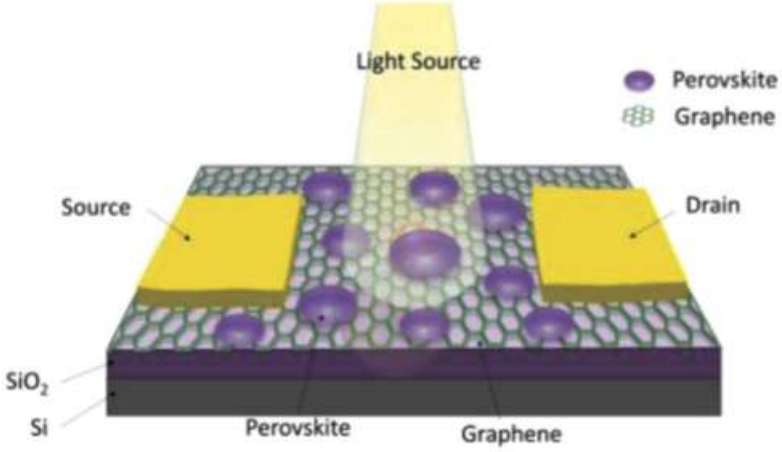


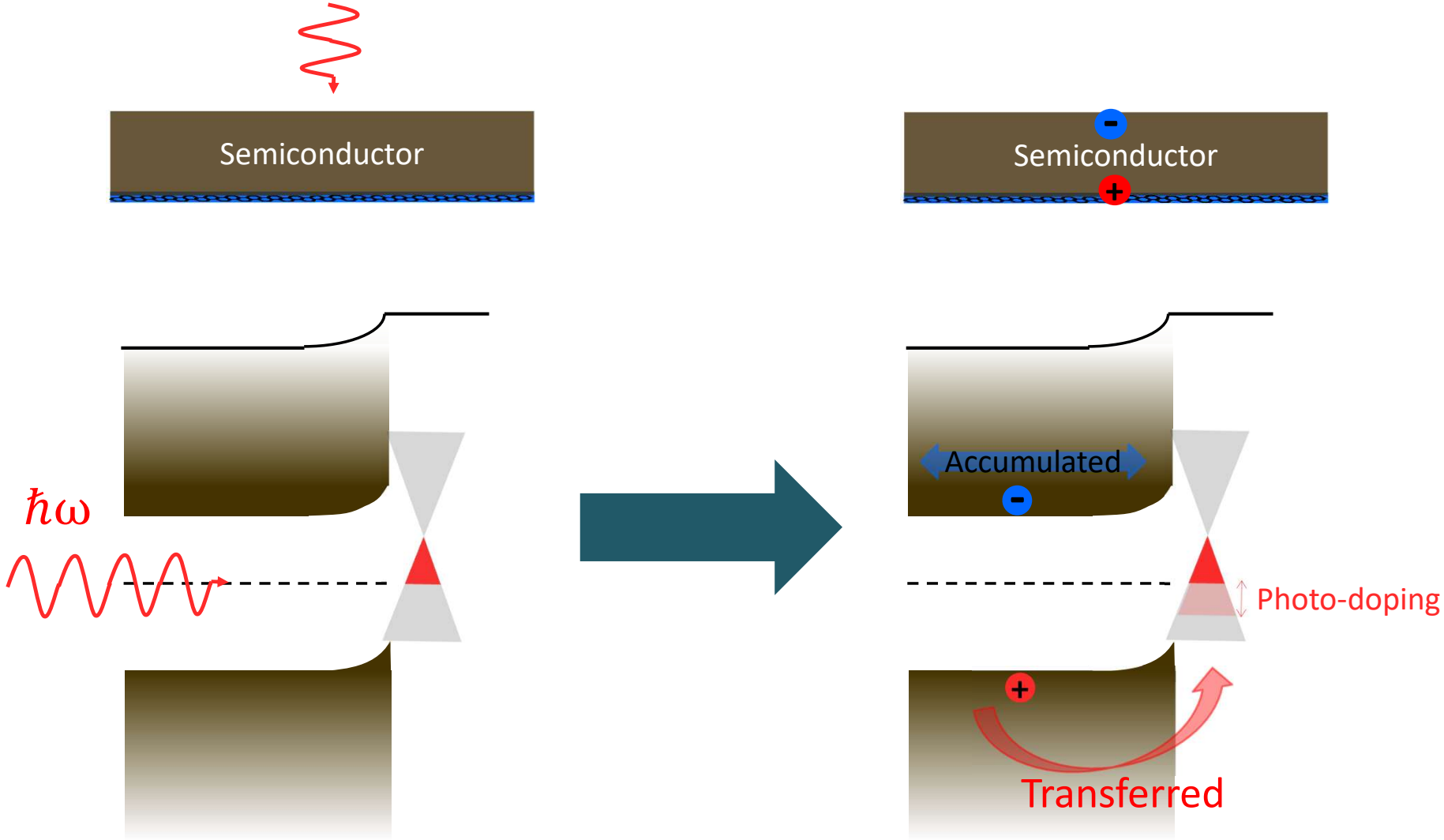
Photo doping

Pros: **Dynamic control**
Reconfigurable doping
Cons: Yet to be discovered



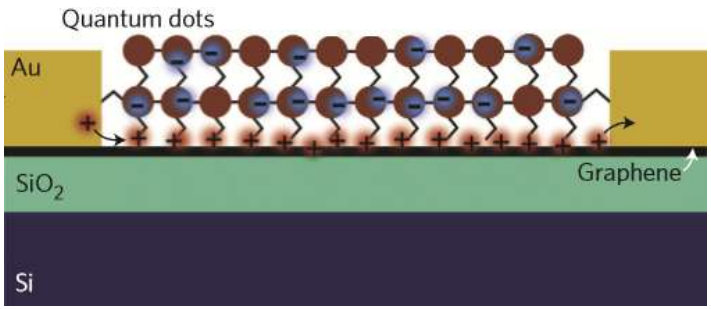
H. Liu et al., *J. Mater. Chem.*, 2011, 21, 3335-3345
P. Qin et al., *Nature communications,* 2014, 5, 3834

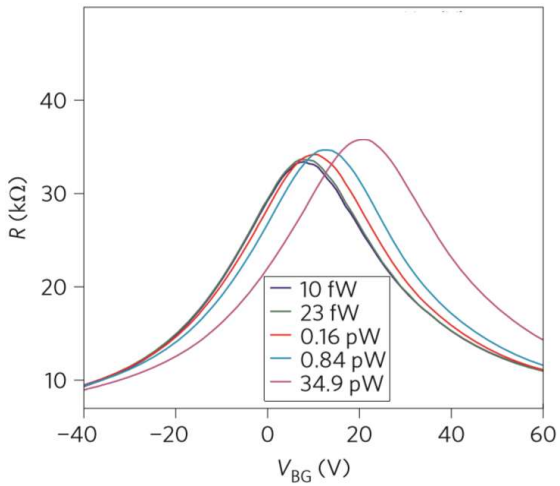
Mechanism of Photo-Doping



Previous Works: Photo-Doping of Graphene

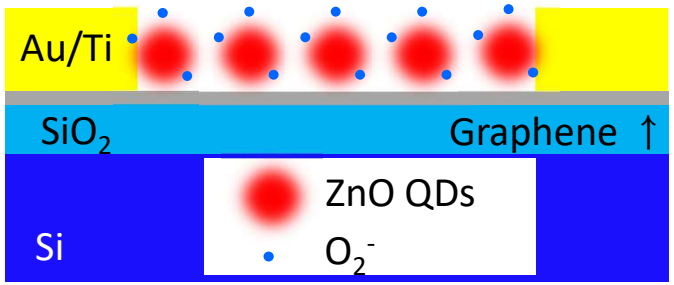
PbS quantum dots

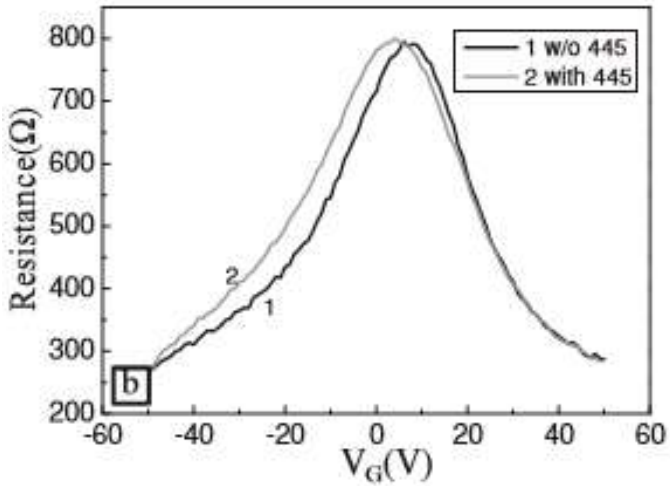




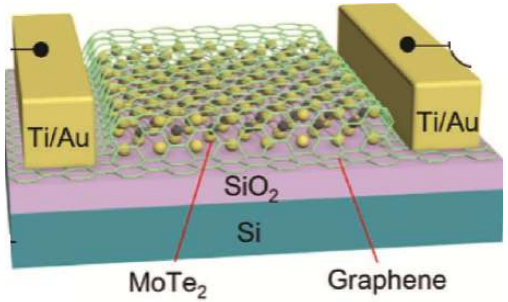
G. Konstantatos et al., *Nature nanotechnology,* 2012, 7.6, 363
W. Guo et al., *Small,* 2013, 9.18, 3031-3036
W. Yu et al., *Small,* 2017, 13.24

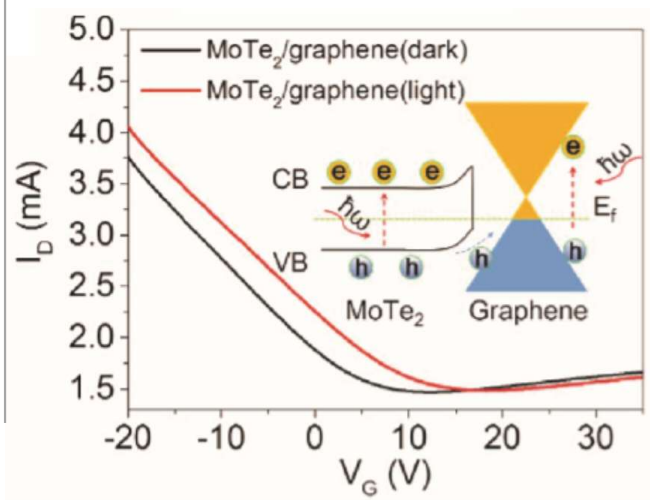
ZnO quantum dots





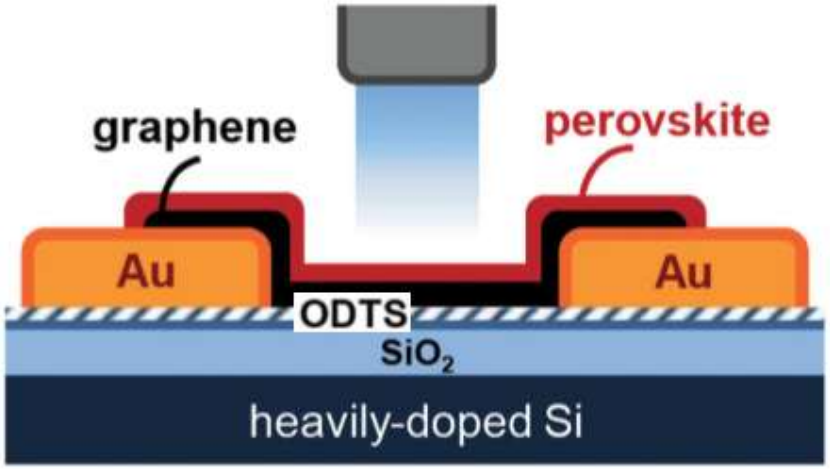
$MoTe_2$



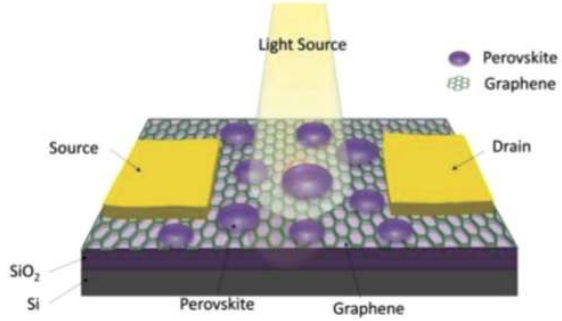


Previous Studies: Perovskite-Graphene Photo-Doping

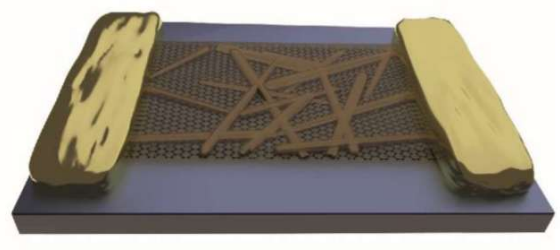
Perovskite film(Lee, 2015)



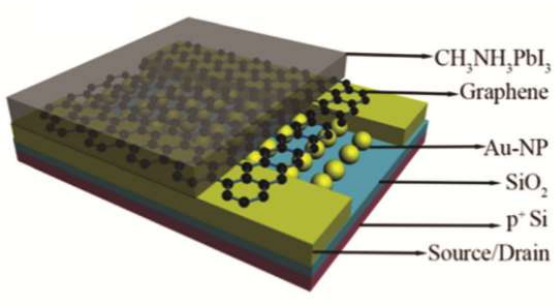
Perovskite Island(Wang, 2015)



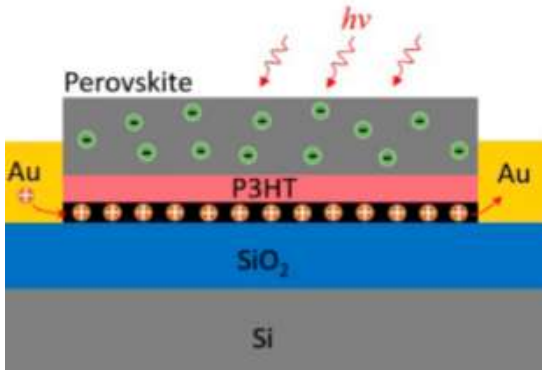
Perovskite NW(Spina, 2015)



Au NPs(Sun, 2016)



P3HT HTL(Xie, 2017)

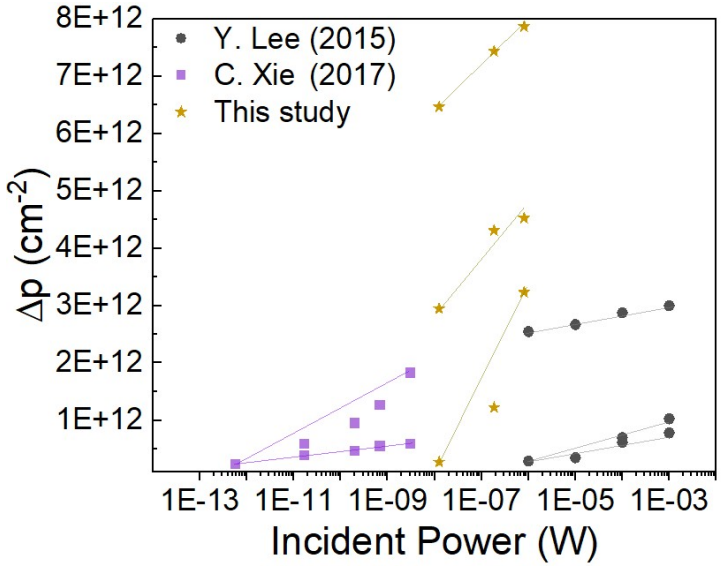


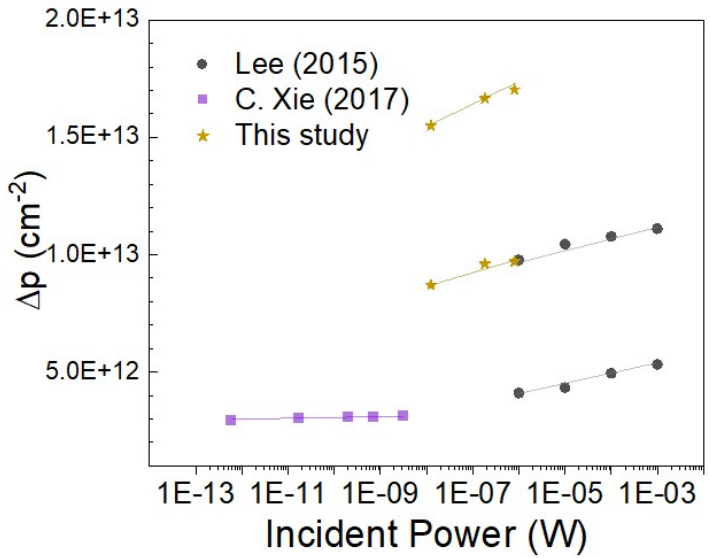
· Focused on responsivity only

· No discussion on doping level

Y. Lee et al., Advanced materials, 2015, 27, 41-46.
Y. Wang et al., *Advanced Optical Materials*, 2015, 3, 1389-1396.
M. Spina et al., *Small*, 2015, 11, 4824-4828.
Z. Sun et al., *Nanoscale*, 2016, 8, 7377-7383.
C. Xie and F. Yan, *ACS applied materials & interfaces*, 2017, 9, 1569-1576.

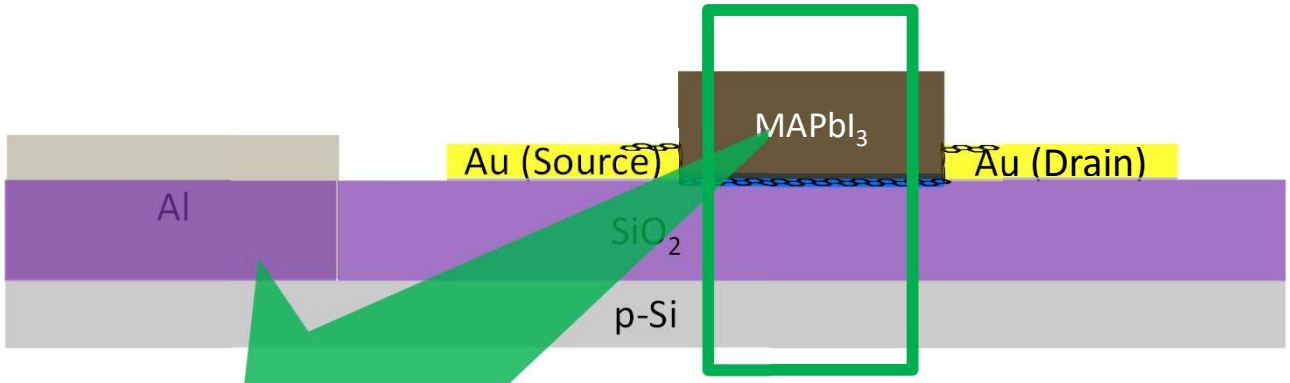
"Enhanced Photo-Doping of Graphene at Reproducible Perovskite-Graphene Heterostructure."

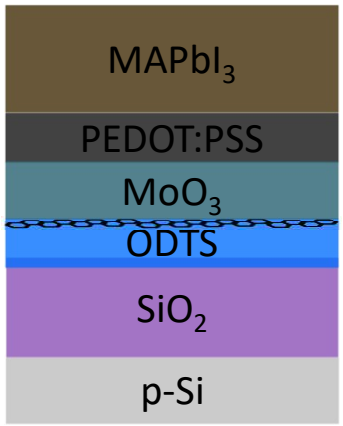


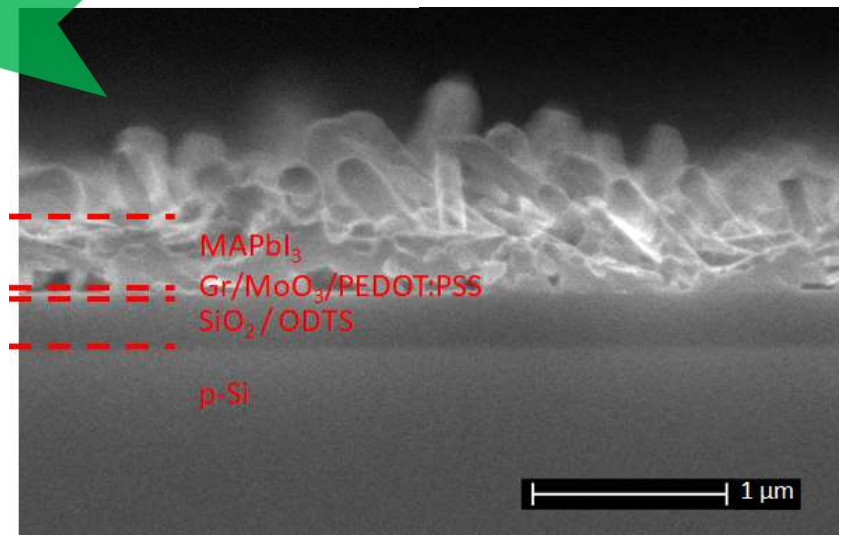


We demonstrate more than 50% increase of carrier density variation compared to previous studies

Device Structure

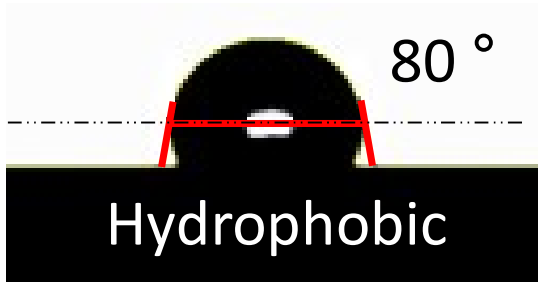


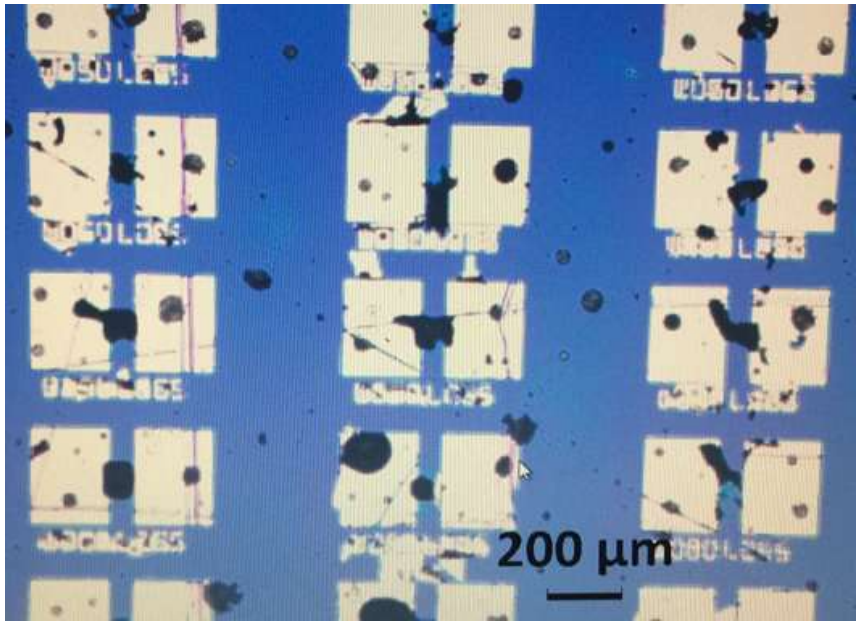




- p-type {111} single crystal Si
- 300nm SiO_2
- 10mM ODTS
- Wet transferred graphene
- 2nm MoO_3
- 30nm PEDOT:PSS
- 400nm $MAPbI_3$

Graphene: Water-Proof Surface



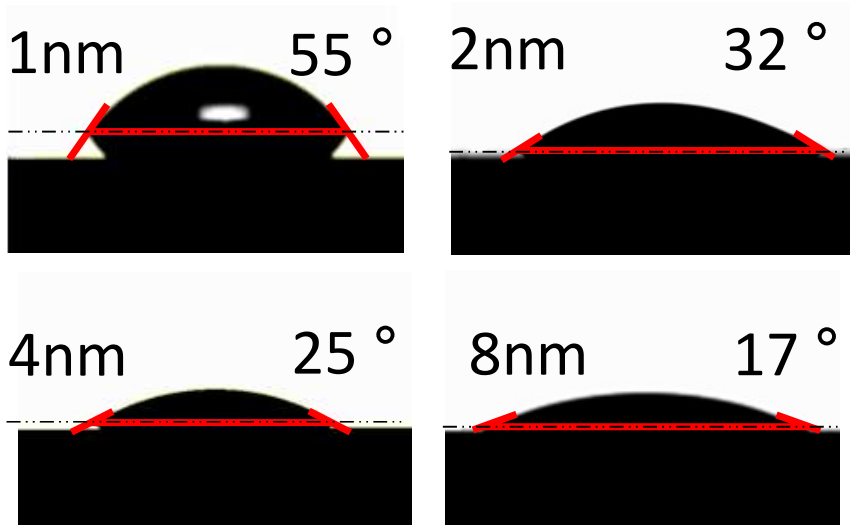


Solution Process on Graphene

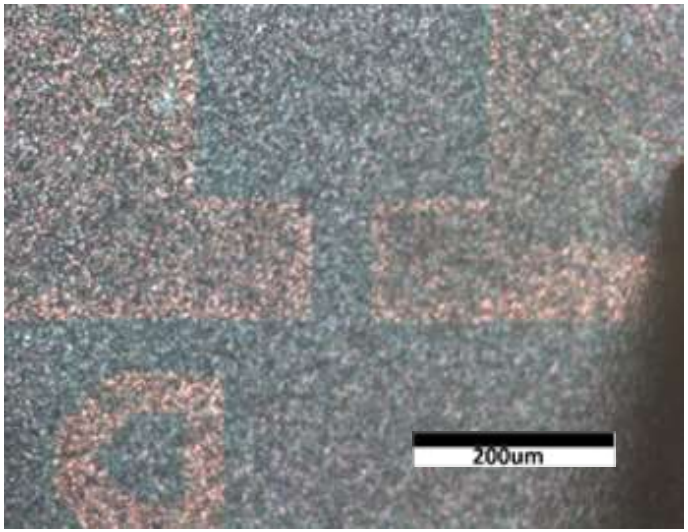
Lee et al., *Nano letters,* 2010, 10.2, 490-493
M. Annamalai et al., *Nanoscale,* 2016. 8.10, 5764-5770

Reproducible: Improved Wettability

MoO_3 Deposition



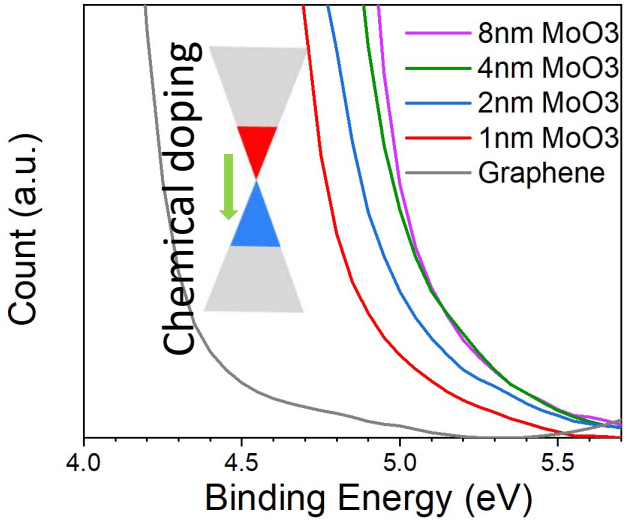
Solution Process on Graphene/MoO_3

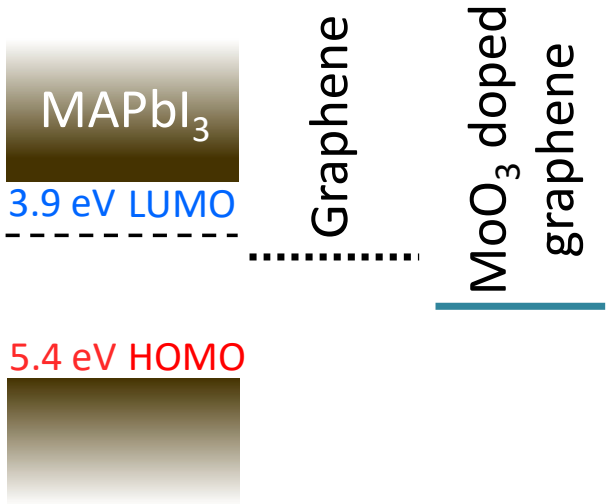


M. Annamalai et al., *Nanoscale*,2016, 8.10, 5764-5770
Sung et al., *Advanced Energy Materials*, 2016, 6.3

MoO_3: Improved Band Alignment

<Fermi-level measurement>





MoO_3 effect

Pros:

- Wettability ↑
- HOMO level alignment

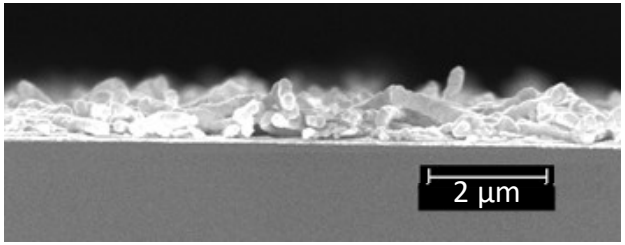
Cons:

- Trap sites ↑
- Conductivity ↓

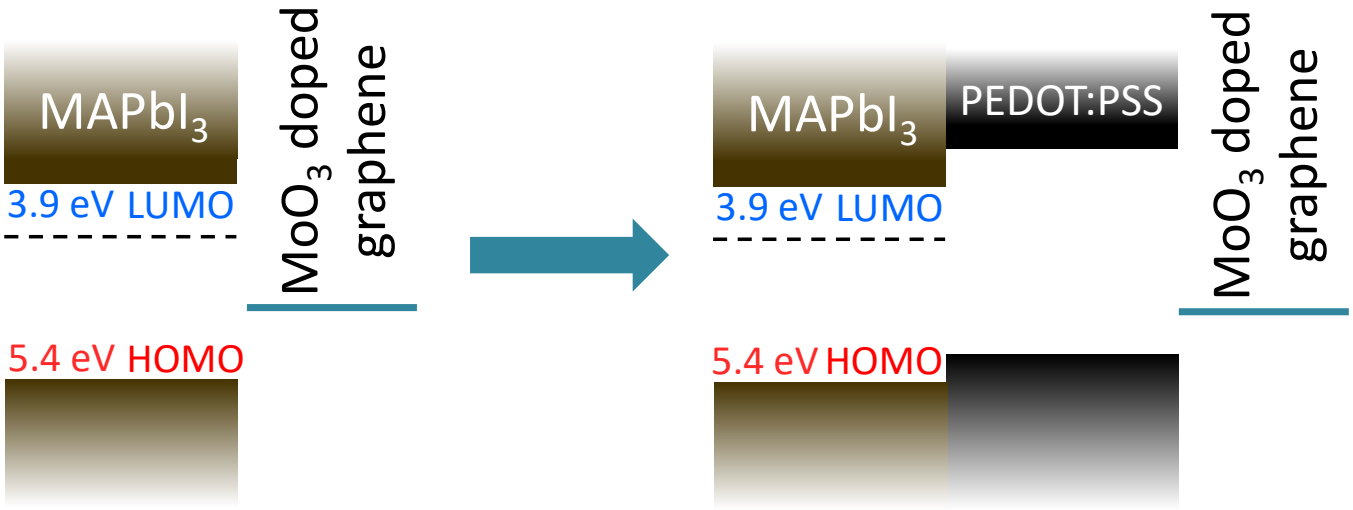
Target MoO_3 Thickness: 2nm

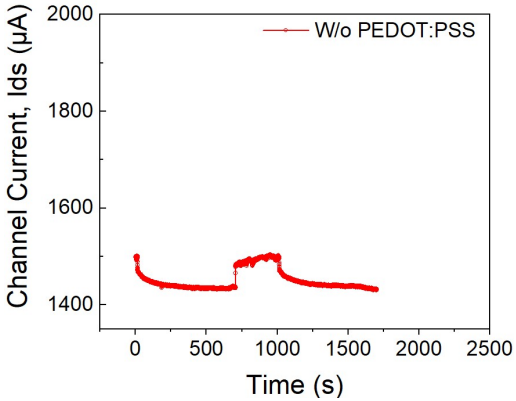
A. Fedorov et al., *APS*, 2010, 3, 46.
H. Liu et al., *J. Mater. Chem.*, 2011, 21, 3335-3345.
D. Wei et al., *Nano Lett.*, 2009, 9 (5), 1752-1758.
Sung et al., *Advanced Energy Materials,* 2016, 6.3

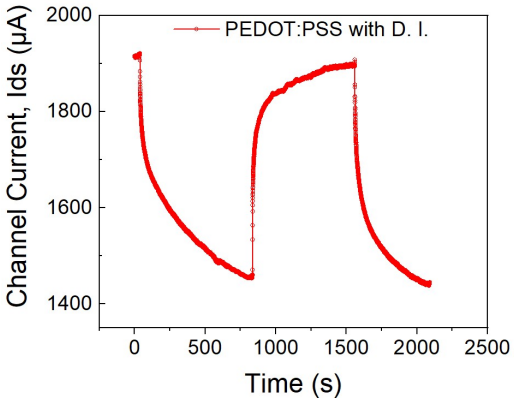
Enhanced Photo-Doping: PEDOT:PSS



Electron blocking layer

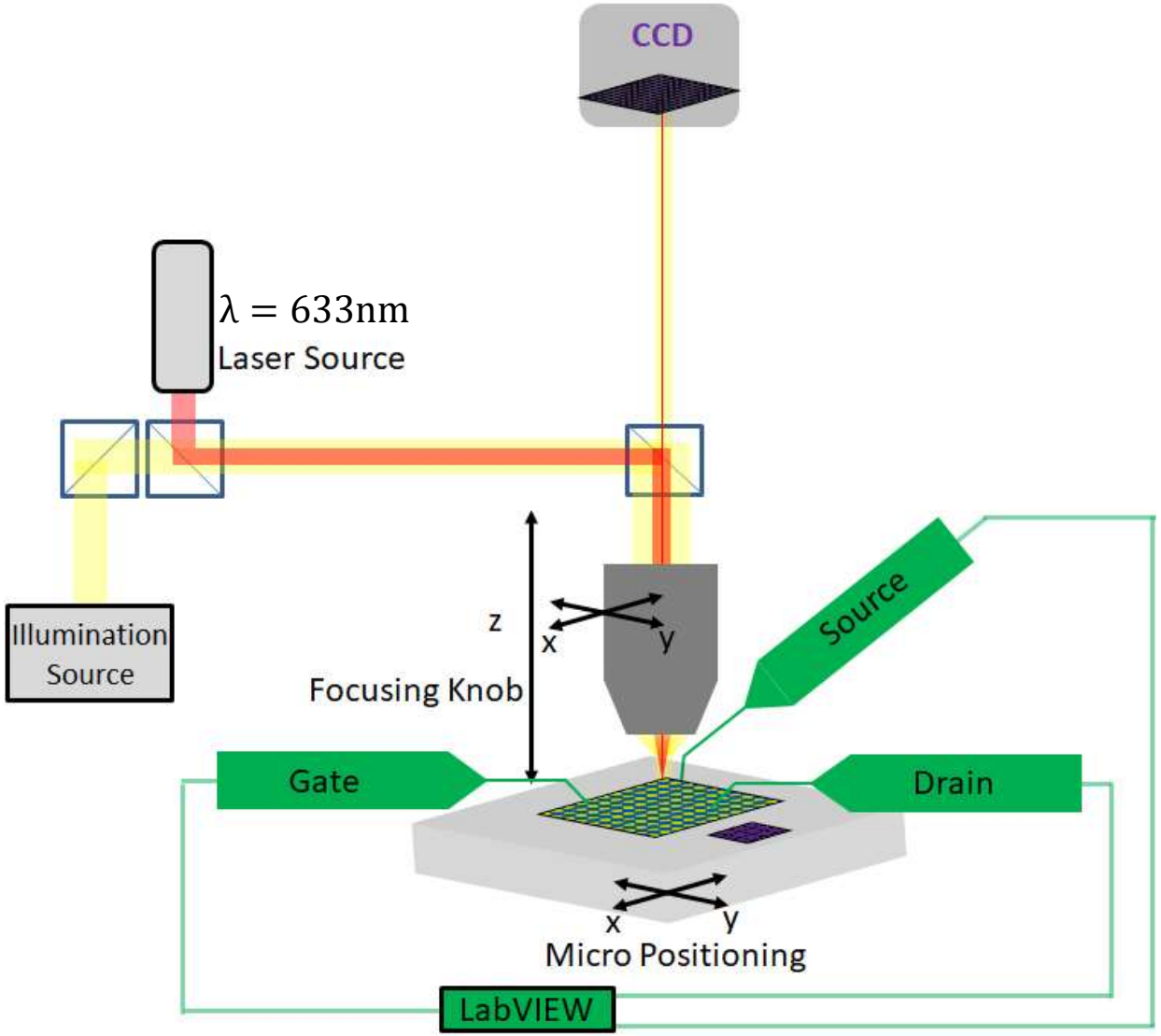




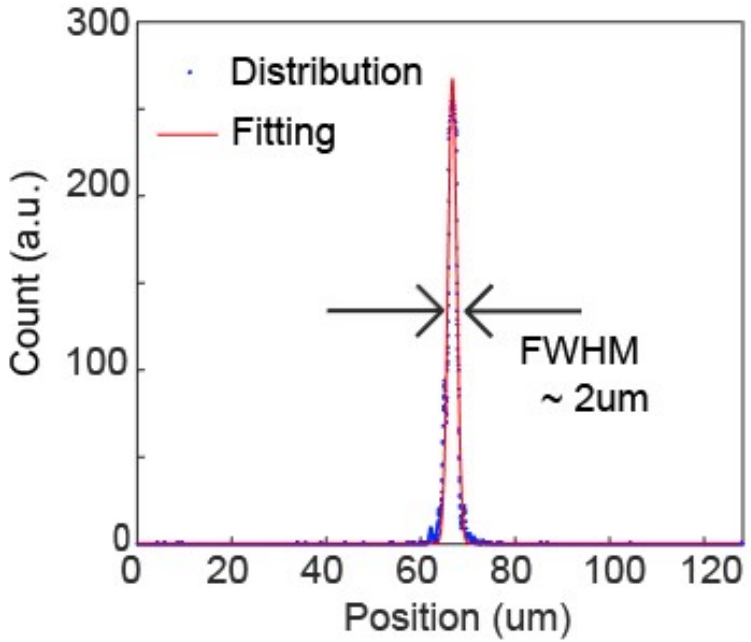


C. Wang et al., *J. Phys. D: Appl. Phys.*, 2018, 51, 025110

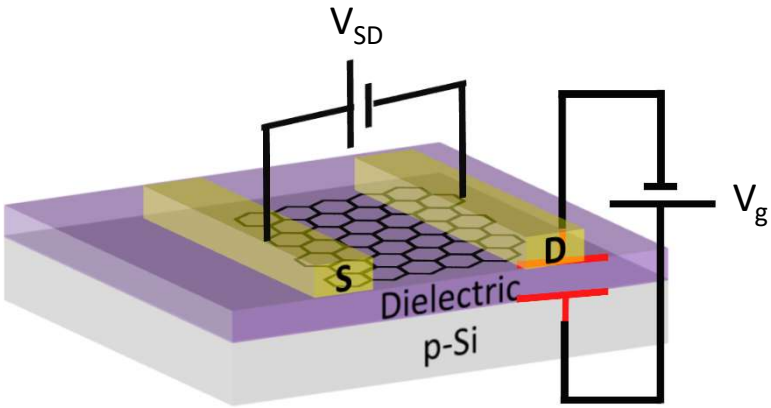
Measuring Set-up

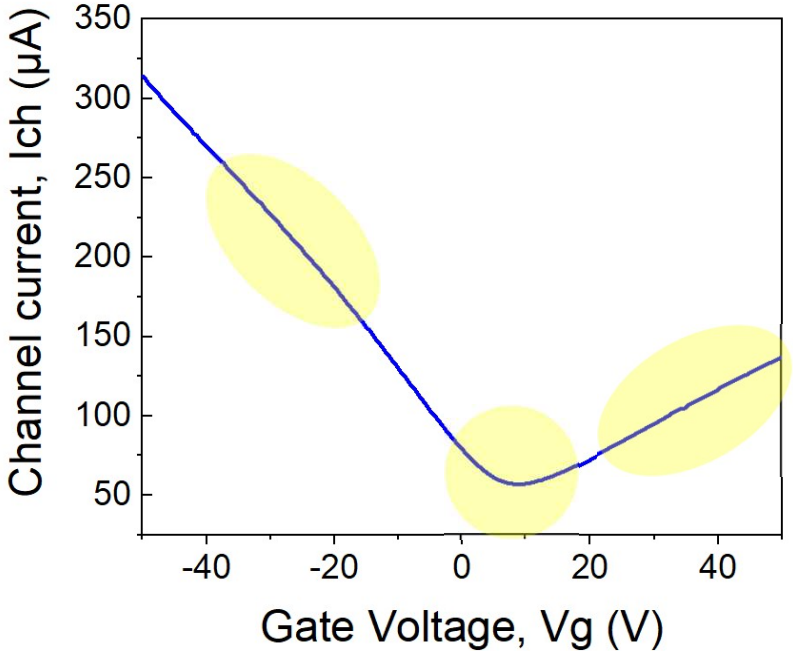


<Beam spot size>



Device Structure: Graphene Field Effect Transistor



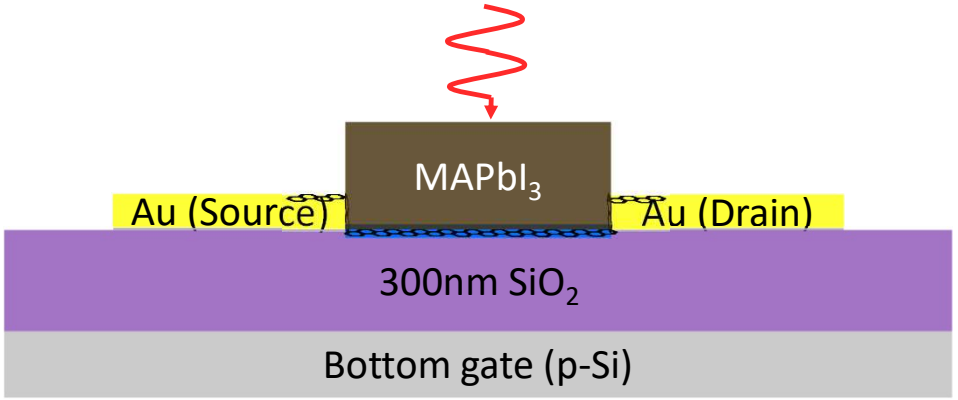


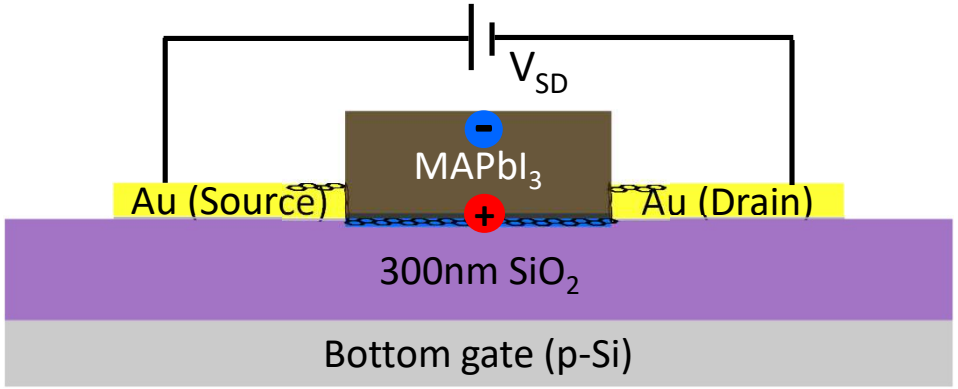
$V_g < V_{dirac}$

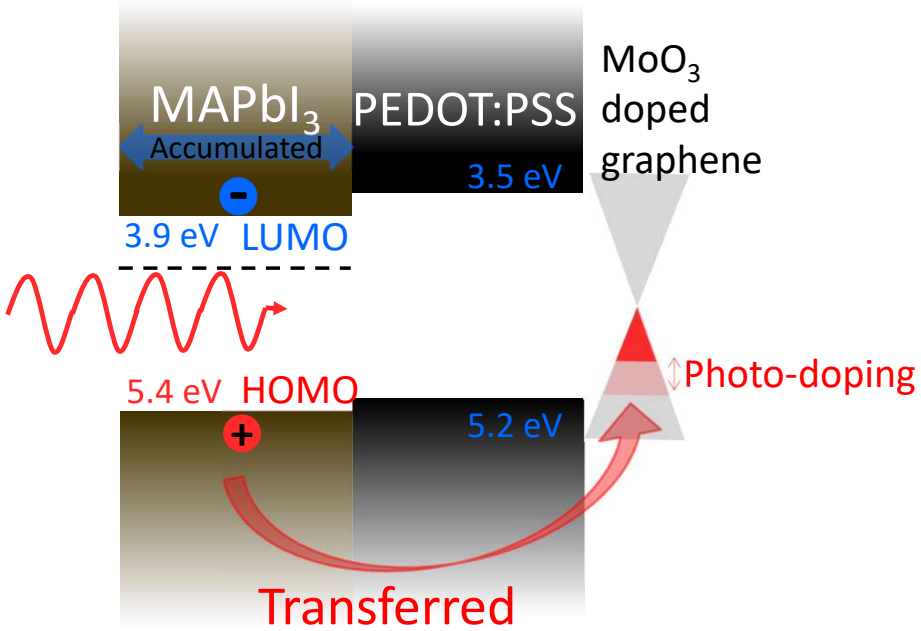
$V_g = V_{dirac}$

$V_g > V_{dirac}$

Measurement: Current Variation under Light

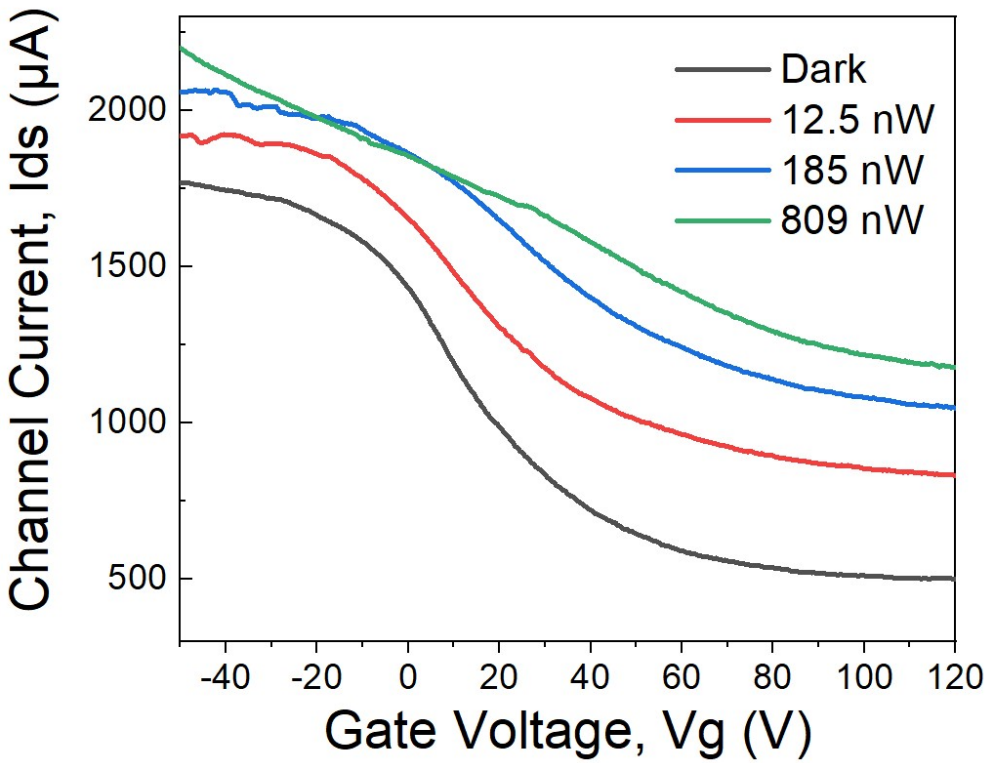






Photocurrent Measurement

Photocurrent measurement under different incident power



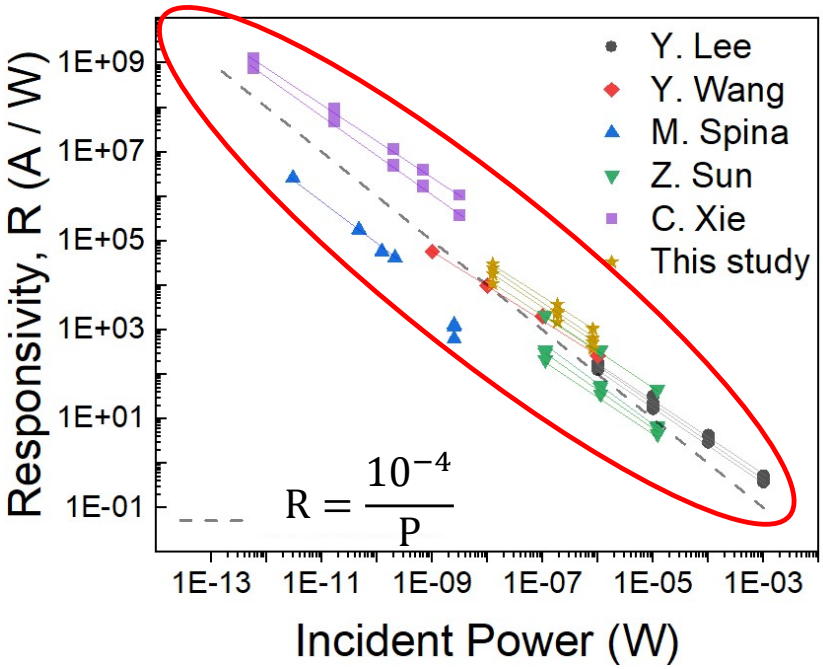
· Responsivity Comparison

· $\Delta p \propto \Delta V_g$

Responsivity Comparison with Previous Studies

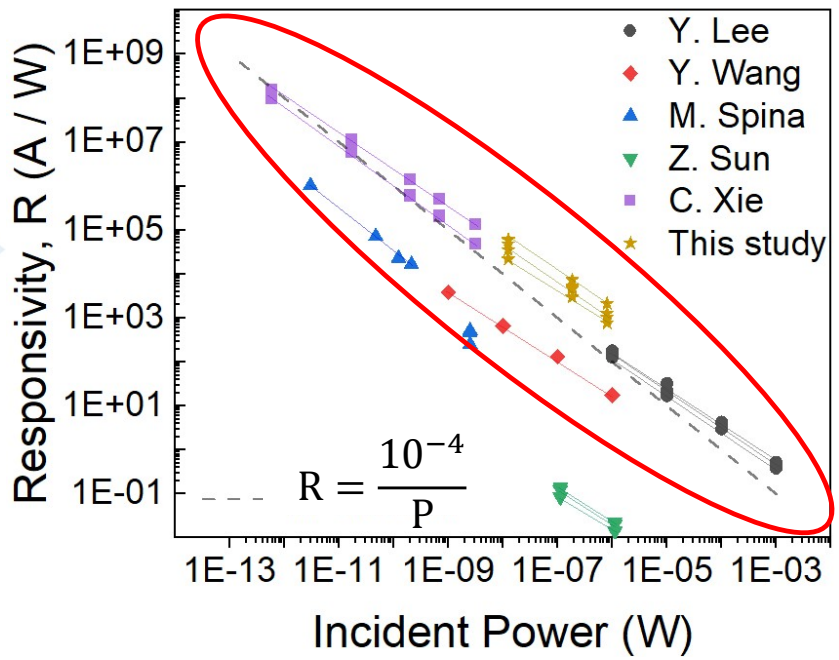
$$R \equiv \frac{I_{ph}}{P_{in}}, \text{ where } I_{ph} \sim e\Delta p\mu V_{SD}\frac{W}{L} \text{ [A]}$$

<Before normalization>



Normalized parameters

<Normalized>

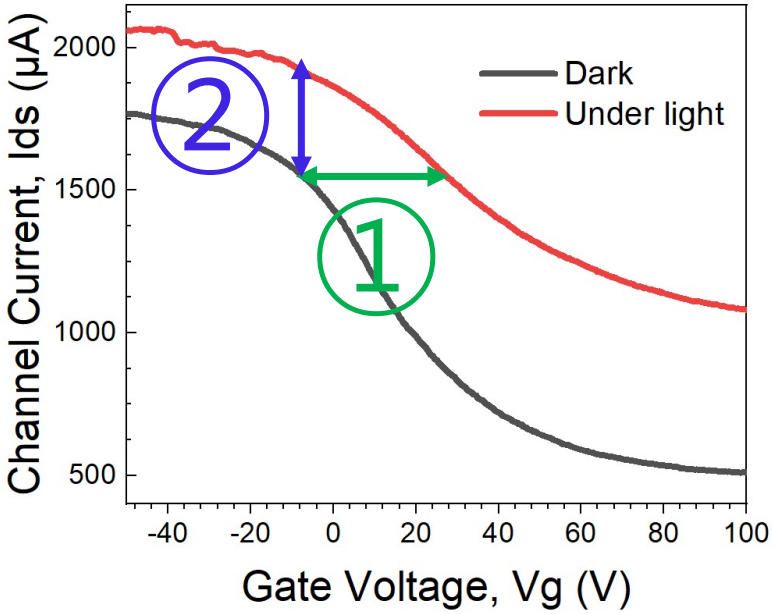


Studied under Photocurrent Saturation Regime

Y. Lee et al., Advanced materials, 2015, 27, 41-46.
Y. Wang et al., *Advanced Optical Materials*, 2015, 3, 1389-1396.
M. Spina et al., *Small*, 2015, 11, 4824-4828.
Z. Sun et al., *Nanoscale*, 2016, 8, 7377-7383.
C. Xie et al., *ACS applied materials & interfaces*, 2017, 9, 1569-1576.

Photo-Doping from Photocurrent Measurement

Calculation of
carrier concentration variation



1

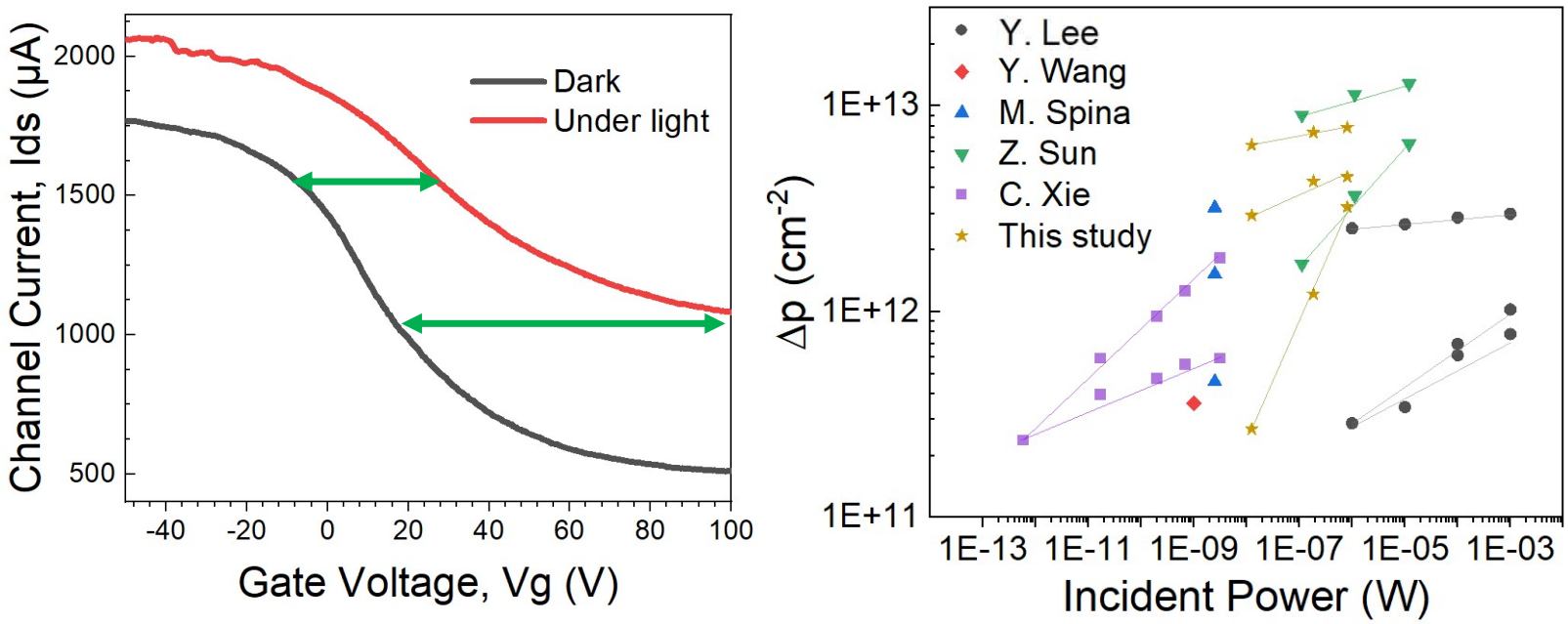
- $Q = CV$ [C]
- $C = \varepsilon \cdot \frac{A}{d}$ [F]
- $\Delta p = \frac{\varepsilon}{ed} \Delta V$ [cm^{-2}]

2

- $I_{dark} = e p_0 \mu V_{SD} \frac{W}{L}$ [A]
- $I_{ph} \sim e \Delta p \mu V_{SD} \frac{W}{L}$ [A]
- $\frac{I_{ph}}{I_{dark}} \sim \frac{\Delta p}{p_0}$
- $p_0 = \frac{\varepsilon}{ed}(V_{dirac} - V_g)$ [cm^{-2}]
- $\Delta p = \frac{I_{ph}}{I_{dark}} \times p_0$ [cm^{-2}]

① Carrier Density Variation from $\triangle V_g$

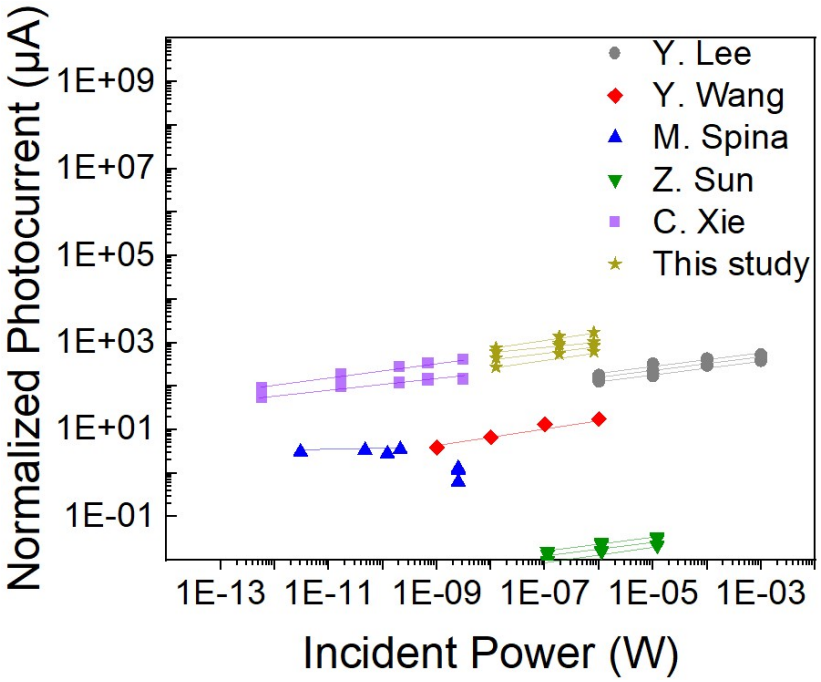
$$① \rightarrow \Delta p = \frac{\varepsilon \Delta V_g}{ed} \quad [\mathrm{cm}^{-2}]$$

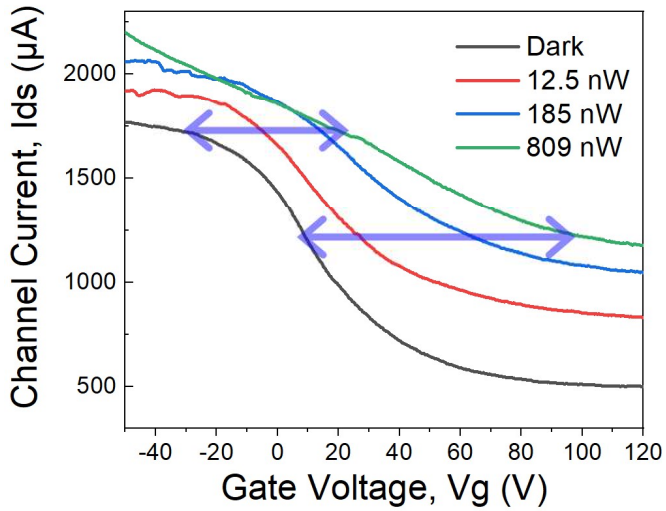


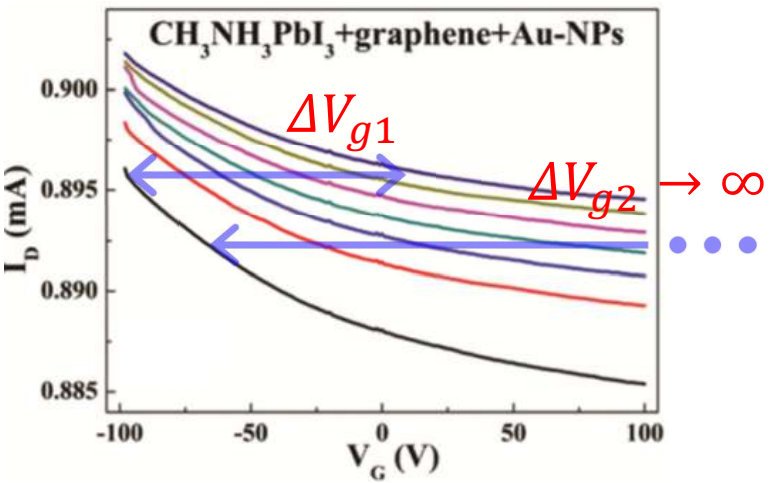
Y. Lee et al., Advanced materials, 2015, 27, 41-46.
Y. Wang et al., *Advanced Optical Materials*, 2015, 3, 1389-1396.
M. Spina et al., *Small*, 2015, 11, 4824-4828.
Z. Sun et al., *Nanoscale*, 2016, 8, 7377-7383.
C. Xie et al., *ACS applied materials & interfaces*, 2017, 9, 1569-1576.

Limitation ①: Low Mobility Makes Large ΔV_g

$$① \rightarrow \Delta p = \frac{\varepsilon \Delta V_g}{ed} \quad [\mathrm{cm}^{-2}]$$







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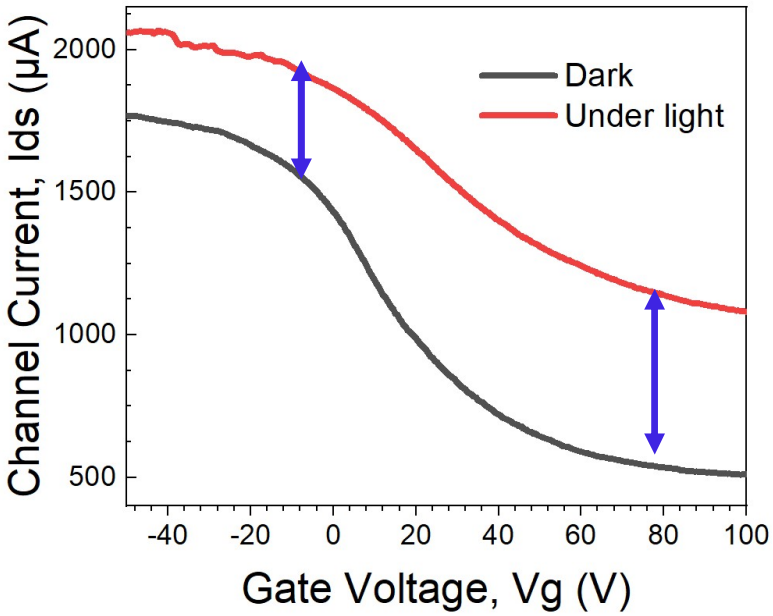
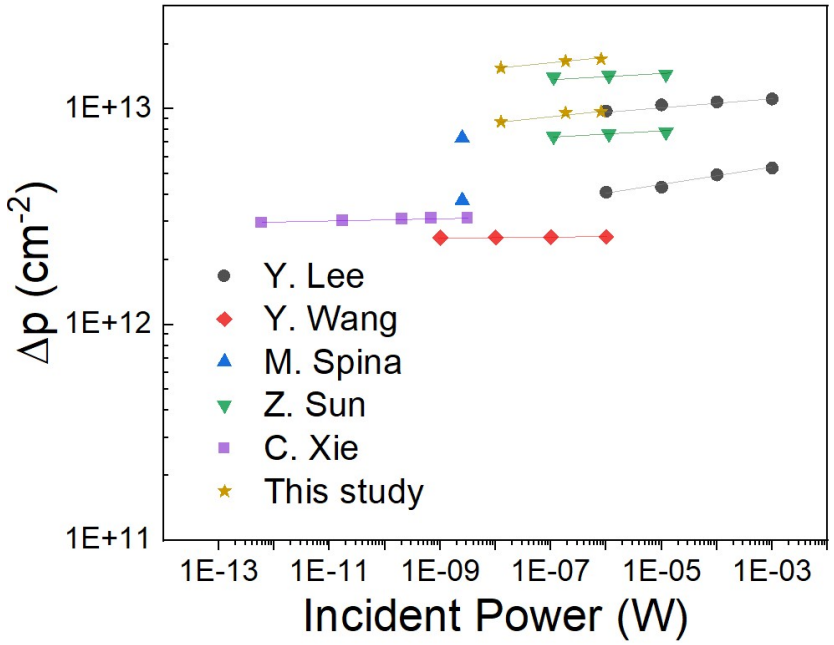
② Carrier Density Variation from I_{ph} / I_{dark}

$$② \rightarrow \Delta p = \frac{I_{ph}}{I_{dark}} \times \frac{\varepsilon \Delta V_g}{ed} \; [\mathrm{cm}^{-2}]$$





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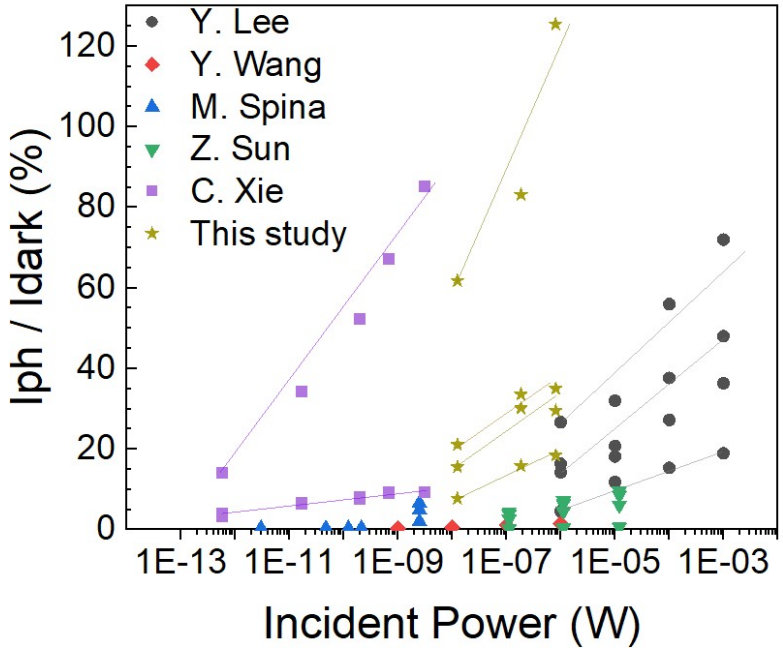
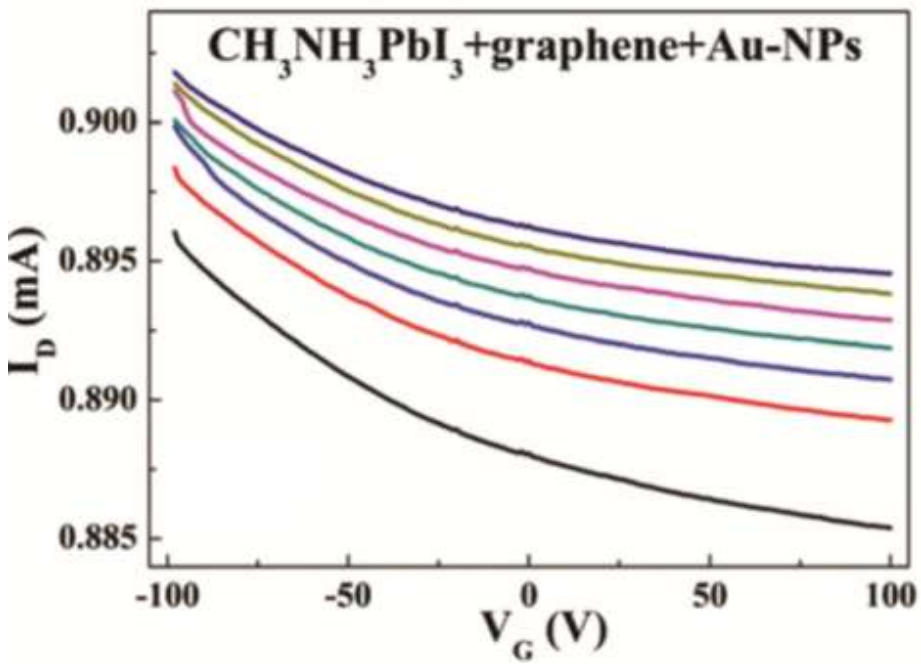
Limitation ②: No Dirac Point Observation

$$② \rightarrow \Delta p = \frac{I_{ph}}{I_{dark}} \times \frac{\varepsilon \Delta V_g}{ed} \quad [\mathrm{cm}^{-2}]$$





$$p_0 = \frac{\varepsilon}{ed}(V_{dirac} - V_g) \quad [\mathrm{cm}^{-2}]$$

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M. Spina et al., *Small*, 2015, 11, 4824-4828.
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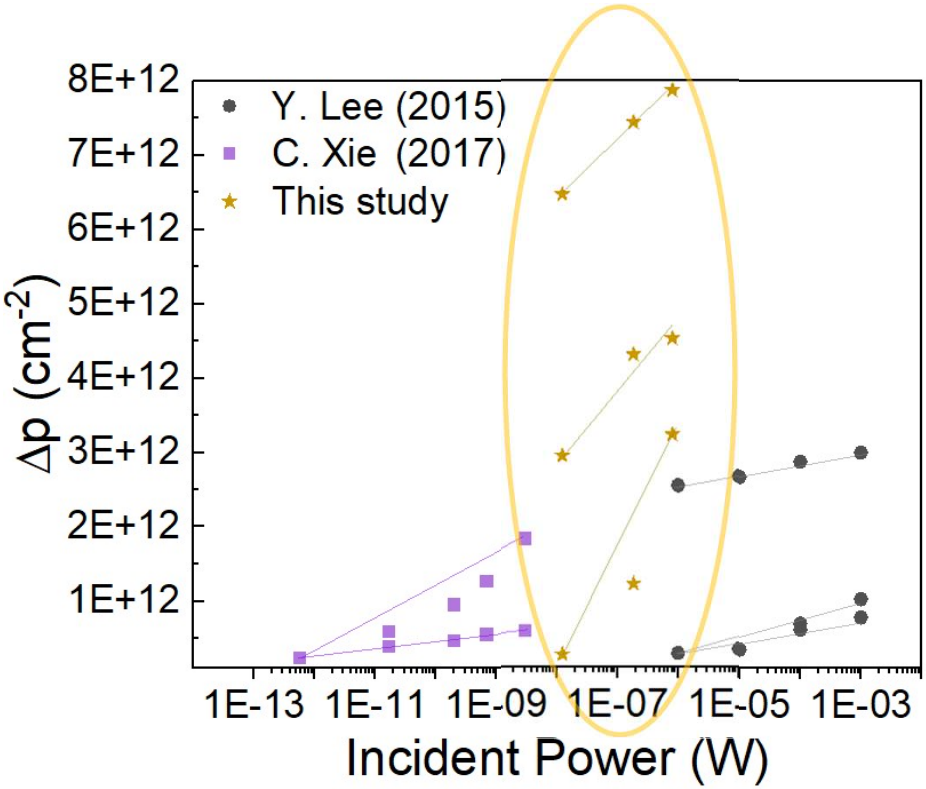
Conclusion



We demonstrate more than 50% increase of carrier density variation compared to previous studies

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